National
Qualifications
2026

X819/76/11 Design and Manufacture

WEDNESDAY, 13 MAY
1:00 PM – 3:15 PM

Total marks — 80

SECTION 1 — 25 marks

Attempt ALL questions.

SECTION 2 — 55 marks

Attempt ALL questions.

Write your answers clearly in the answer booklet provided. In the answer booklet you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

You must leave your answer booklet on your desk; if you do not, you could lose all the marks for this paper.

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SECTION 1 — 25 marks

Attempt ALL questions

1. Two lamps are shown below with product information.

Adjustable tripod spotlight

Materials

stained solid beech frame

stainless steel spotlight casing

acrylic spotlight lens

ABS on/off control switch

brass fixings

Additional details

foot operated on/off switch









Dimmable uplighter with reading lamp

Materials

chrome plated tubular mild steel frame

chrome plated mild steel shades

mild steel base

Additional details

dimmable light

adjustable reading lamp

self–assembly

MARKS

1. (continued)

(a) Explain why the materials chosen are suitable for the lamps. **6**

(*You must give **six** different explanations*.)

(b) Name **three** appropriate manufacturing processes used in the production of the lamps **and** explain why each one is suitable. **6**

(c) Describe how function has influenced the design of the lamps. **5**

(d) Describe how different aesthetic aspects have influenced the design of the lamps. **4**

(You must refer to a range of aesthetic aspects in your answer.)

(e) Describe how production and planning systems could be used to improve efficiency during the manufacture of products such as the lamps. **4**

[Turn over

MARKS

SECTION 2 — 55 marks

Attempt ALL questions

2. A badminton racket is shown below.



The frame of the badminton racket is manufactured from a composite material.

(a) State the name of a suitable composite material **and** explain the benefits of using composite materials in the manufacture of products such as the badminton racket. **3**

Different techniques can be used to evaluate the performance of the badminton racket.

(b) Describe the key stages of the following techniques that can be used to evaluate performance:

- user trial
- test rigs. **4**

Branding can influence the success of a product.

(c) Describe the benefits of launching under a successful brand name. **2**

MARKS

3. Companies protect their Intellectual Property Rights (IPR) using a variety of methods.

(a) Identify **two** methods of protecting the IPR of products **and** outline what each method protects. **4**

As part of their role within the design team, lawyers are responsible for securing IPR.

(b) Describe the role of **two** other members of a design team. **4**

4. A hairdryer is shown below.



The casing for the hairdryer has been manufactured using ABS.

(a) Explain why ABS is suitable for the manufacture of the casing of this product. **2**

The designer may have used brainstorming to generate ideas for the hairdryer.

(b) Explain the benefits of using brainstorming to generate ideas. **2**

Designers would have used a variety of graphic techniques throughout the design of the hairdryer.

(c) Describe how graphic techniques could have been used to communicate during the following stages of the design process:

- initial ideas
- manufacture
- marketing. **6**

[Turn over

MARKS

5. A robot vacuum cleaner is shown below.



The robot vacuum cleaner has been targeted at a niche market.

(a) Explain what is meant by the term niche market. **2**

Technology push and market pull can result in new or improved products being created, like the robot vacuum cleaner.

(b) Describe how technology push **and** market pull have influenced the design of a product(s) with which you are familiar. **4**

Computer Aided Design (CAD) may have been used during the development of the robot vacuum cleaner.

(c) Explain the advantages of using CAD during the development of products such as the robot vacuum cleaner. **4**

MARKS

6. Symbols and labels can be included on products to help identify materials.

(a) Describe methods, other than symbols and labels, that could be used to identify woods, metals and/or plastics. **3**

Products can be assembled using knock–down fittings.

(b) Explain the advantages of using knock–down fittings in the assembly of products. **3**

Many products are now manufactured using fully automated production methods.

(c) Explain the impact of fully automated production methods on the workforce and society. **4**

7. Successful products will have had careful consideration of all three areas of ergonomics.

Describe how ergonomics has influenced the design of products with which you are familiar. **8**

(*You should refer to all three areas of ergonomics in your answer*.)

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Question 1 Adjustable tripod — Three images of Two Tone Spotlight Floor Lamp with Wooden Tripod are taken from https://www.culinaryconcepts.co.uk/spotlight-floor-lamp-with-wood-tripod?srsltid=AfmBOoru12sI6PZ2paWE_wjZvrm-PP0h5rS3MujeDegtuHHqqnQohOs%7C%20.

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Dimmable uplighter — Three images of Abasíes 2 Light Satin Chrome Mother & Child Floor Lamp With Dimmable Uplighter & Dimmable, Adjustable Reading Lamp are taken from https://www.thelightingcentre.co.uk/lnmo4n4946-abas-es-2-light-satin-chrome-mother-child-floor-lamp-with-dimmable-uplighter-dimmable-adjustable-reading-lamp.html.

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Question 2 Image of Adidas Spieler e activ badminton racket.

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